Dataset 1: Penguins – EDA

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set(style="whitegrid")
```

```
url = "https://raw.githubusercontent.com/rerkki/data-analytics-ml/refs/heads
```

```
url = "https://raw.githubusercontent.com/rerkki/data-analytics-ml/refs/heads
df = pd.read_csv(url)
df.head()
```

	species	island	culmen_length_mm	culmen_depth_mm	flipper_length_mm	b
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	

Next steps: Generate code with `df` New interactive sheet

```
df.shape
df.info()
df.isna().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    object
 1   island             344 non-null    object
 2   culmen_length_mm   342 non-null    float64
 3   culmen_depth_mm    342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                334 non-null    object
dtypes: float64(4), object(3)
memory usage: 18.9+ KB
```

	0
species	0
island	0
culmen_length_mm	2
culmen_depth_mm	2
flipper_length_mm	2
body_mass_g	2
sex	10

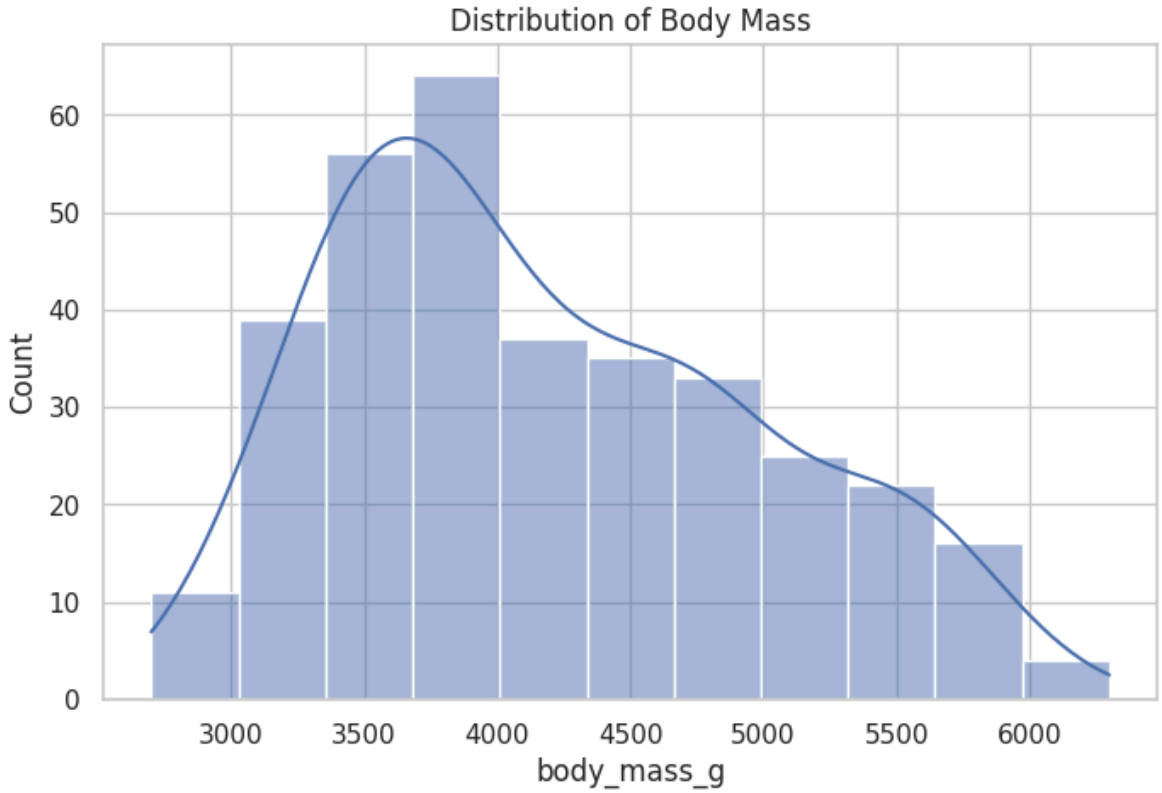
dtype: int64

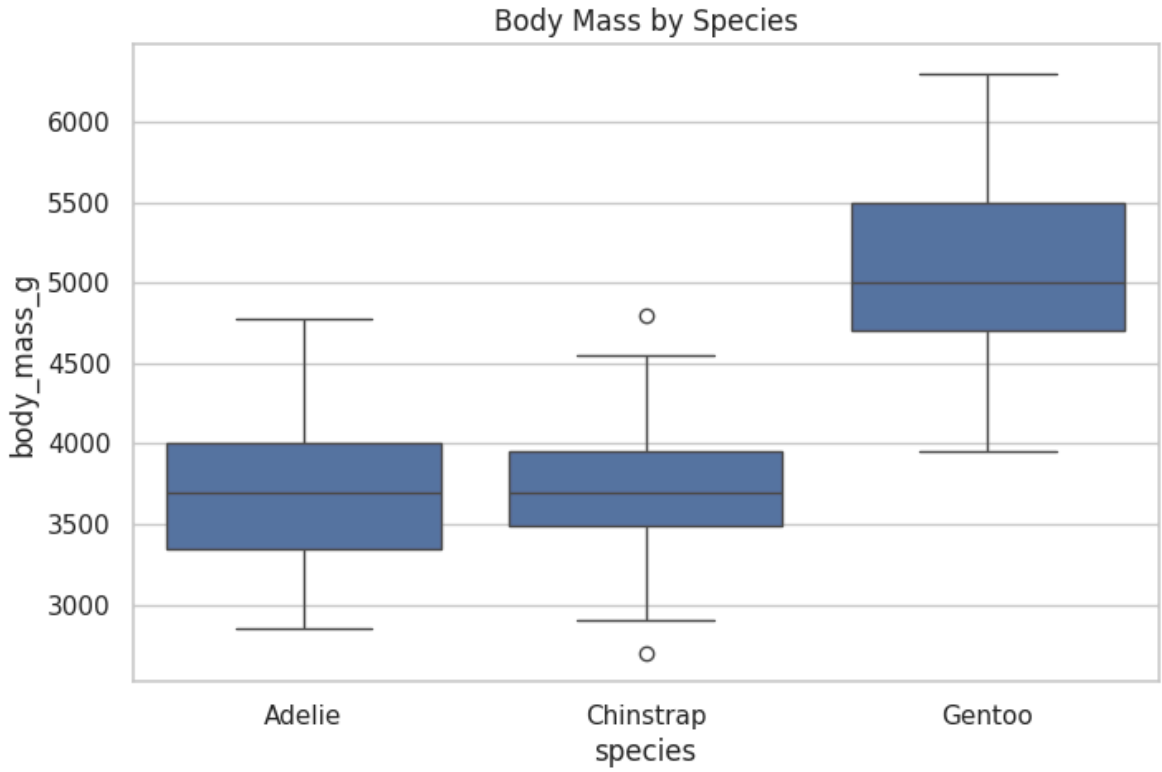
```
df.describe()
```

	culmen_length_mm	culmen_depth_mm	flipper_length_mm	body_mass_g
count	342.000000	342.000000	342.000000	342.000000
mean	43.921930	17.151170	200.915205	4201.754386
std	5.459584	1.974793	14.061714	801.954536
min	32.100000	13.100000	172.000000	2700.000000
25%	39.225000	15.600000	190.000000	3550.000000
50%	44.450000	17.300000	197.000000	4050.000000
75%	48.500000	18.700000	213.000000	4750.000000
max	59.600000	21.500000	231.000000	6300.000000

```
plt.figure(figsize=(8,5))
sns.histplot(df["body_mass_g"], kde=True)
plt.title("Distribution of Body Mass")
plt.show()
```

```
plt.figure(figsize=(8,5))
sns.boxplot(x="species", y="body_mass_g", data=df)
plt.title("Body Mass by Species")
plt.show()
```



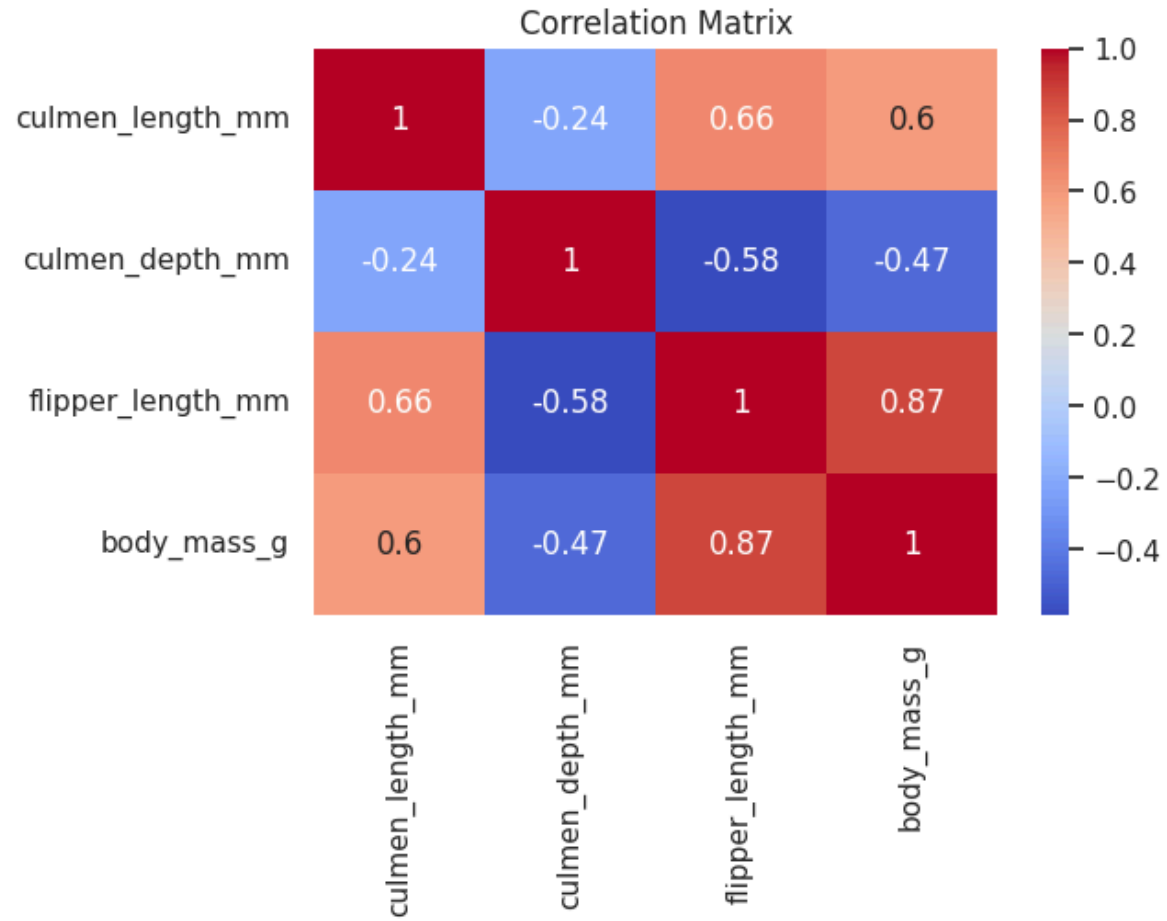


```
df.columns
```

```
Index(['species', 'island', 'culmen_length_mm', 'culmen_depth_mm',
       'flipper_length_mm', 'body_mass_g', 'sex'],
      dtype='object')
```

```
numeric_cols = ["culmen_length_mm", "culmen_depth_mm", "flipper_length_mm",
corr = df[numeric_cols].corr()

plt.figure(figsize=(6,4))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```



In the Penguins dataset, there are three species with clear differences in body mass and flipper length. The distribution of body mass is roughly normal with some outliers. There is a strong positive correlation between flipper length and body mass, meaning larger penguins tend to be heavier.

Dataset 2: Titanic – EDA

```
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/tit
df_titanic = pd.read_csv(url)
df_titanic.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/(31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734

Next steps: Generate code with `df_titanic` New interactive sheet

```
df_titanic.shape
df_titanic.info()
df_titanic.isna().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

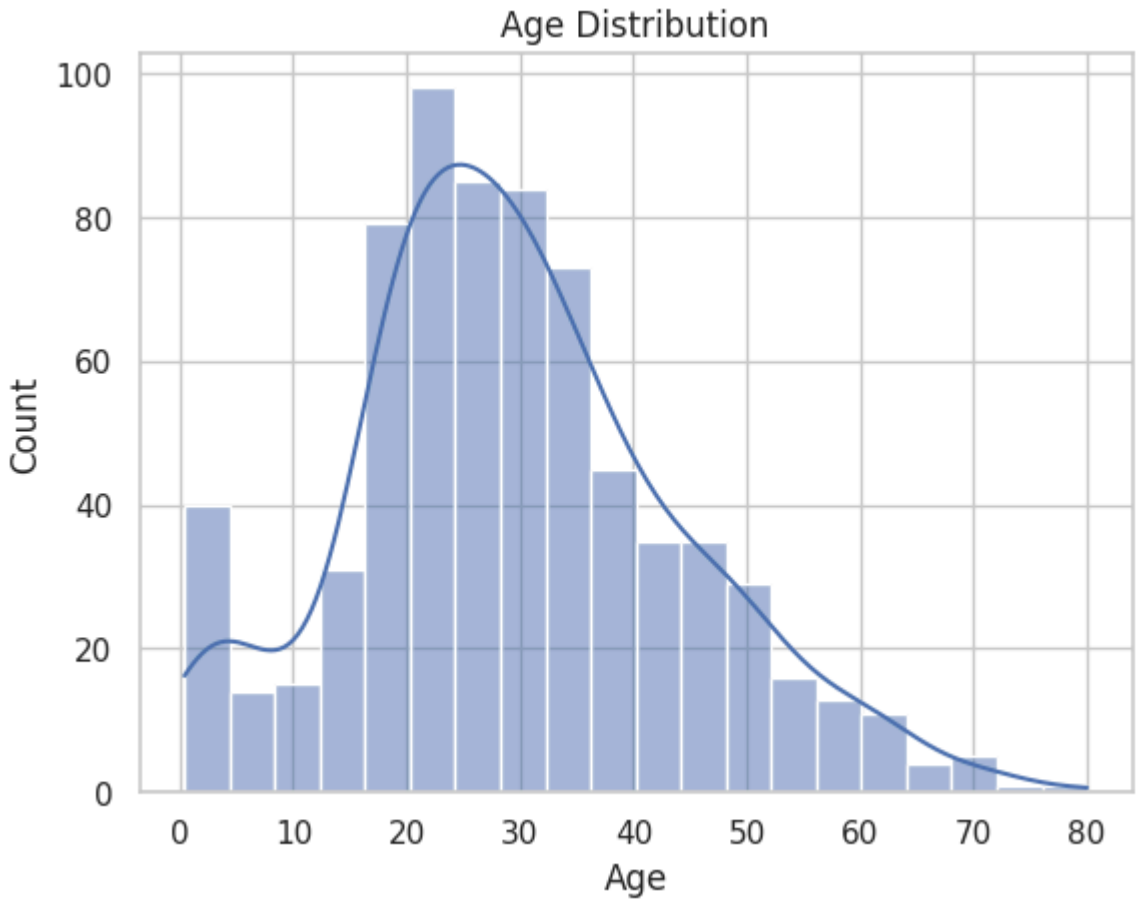
	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

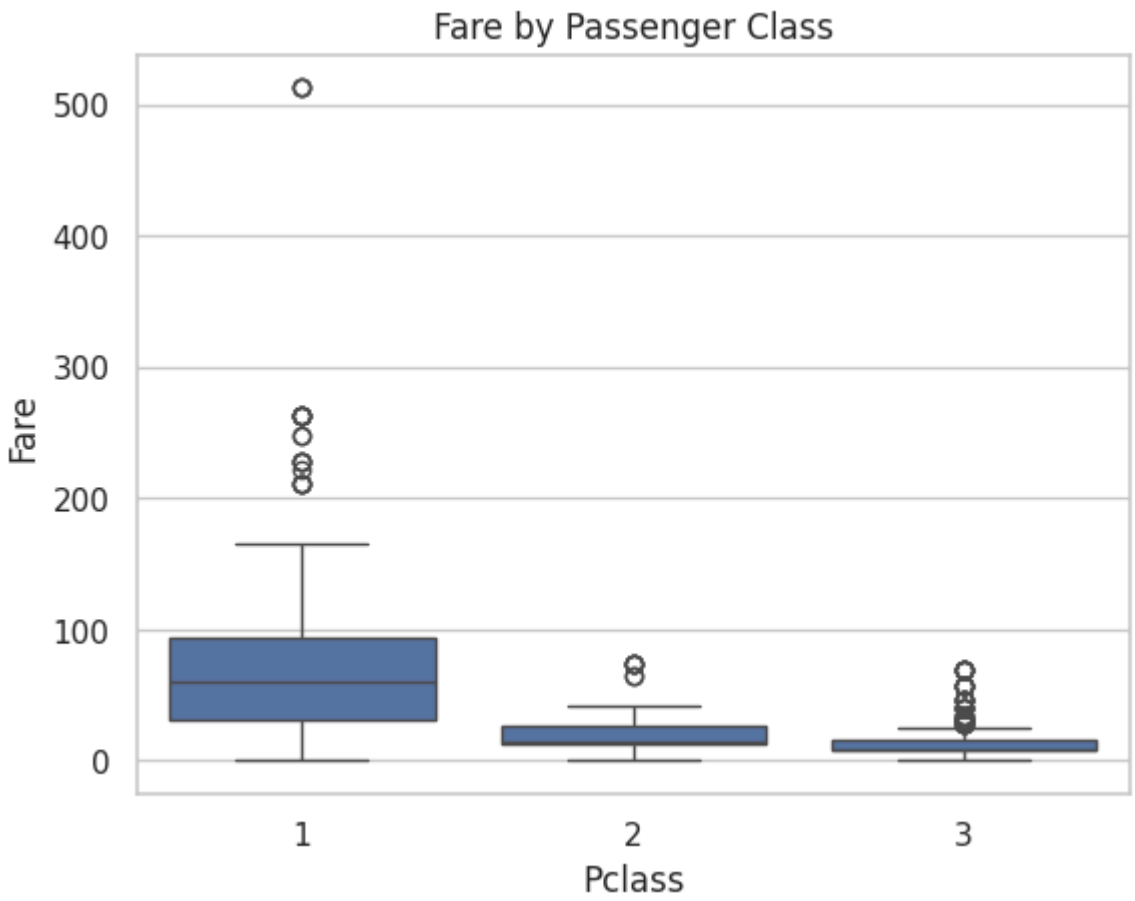
```
df_titanic.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

```
sns.histplot(df_titanic["Age"], kde=True)
plt.title("Age Distribution")
plt.show()
```

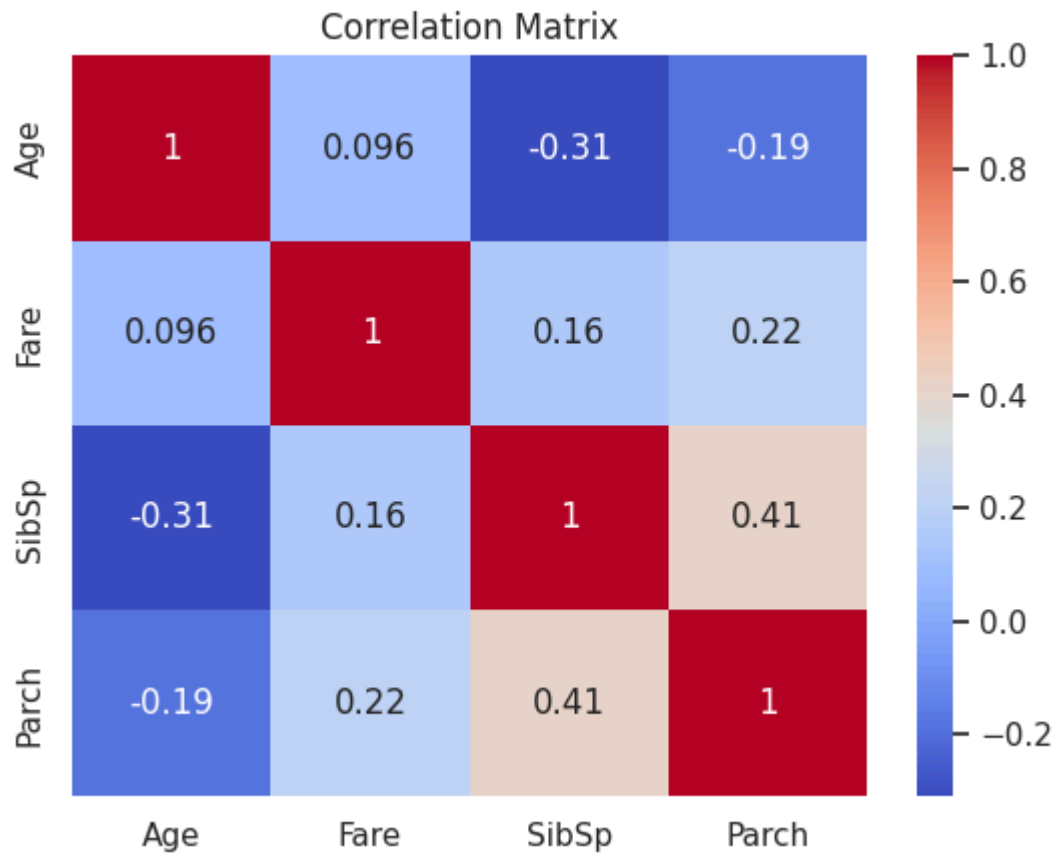



```
sns.boxplot(x="Pclass", y="Fare", data=df_titanic)
plt.title("Fare by Passenger Class")
plt.show()
```




```
numeric_cols = ["Age", "Fare", "SibSp", "Parch"]
corr = df_titanic[numeric_cols].corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```



The Titanic dataset shows clear differences in fare between passenger classes. Age distribution is right-skewed with missing values. Fare has strong positive correlation with passenger class. Some outliers exist in fare values

Dataset 3: Iris – EDA

```
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.cs
df_iris = pd.read_csv(url)
df_iris.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with `df_iris` New interactive sheet

```
df_iris.shape
```

```
(150, 5)
```

```
df_iris.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df_iris.isna().sum()
```

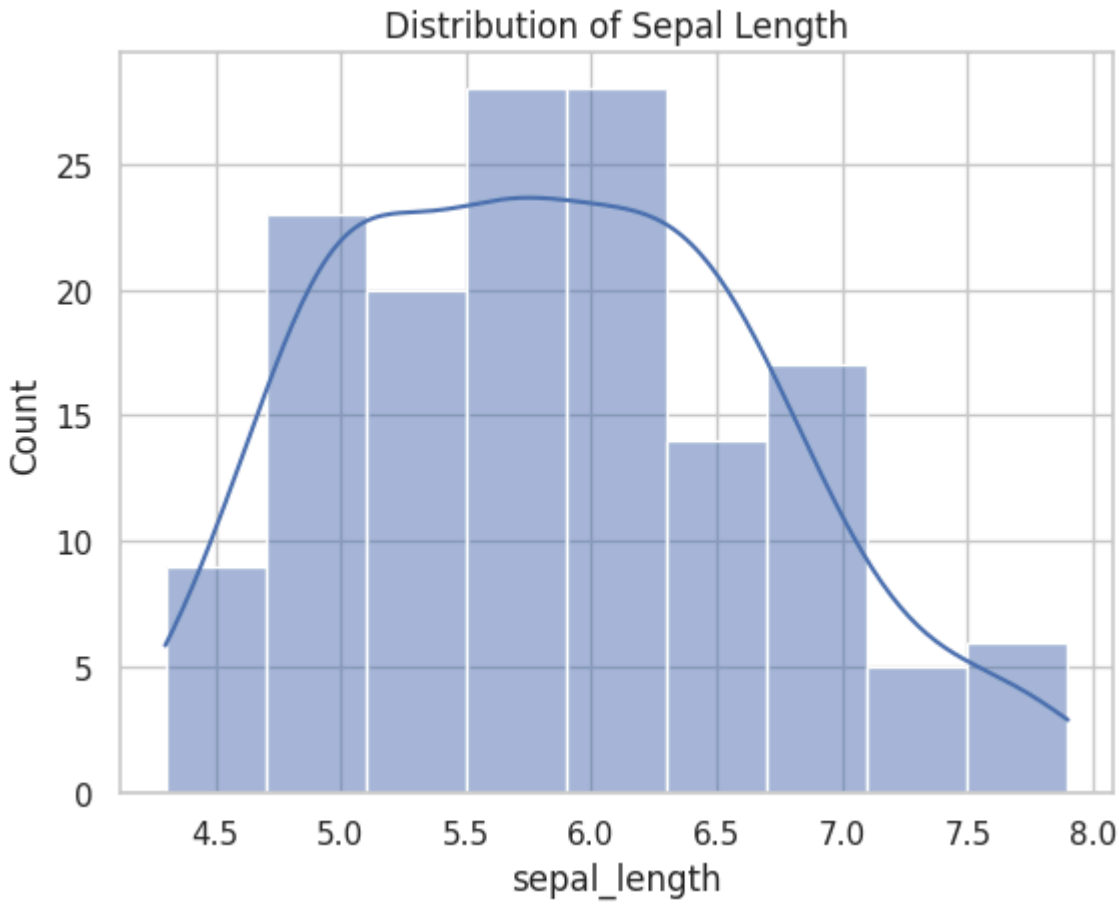
	0
sepal_length	0
sepal_width	0
petal_length	0
petal_width	0
species	0

dtype: int64

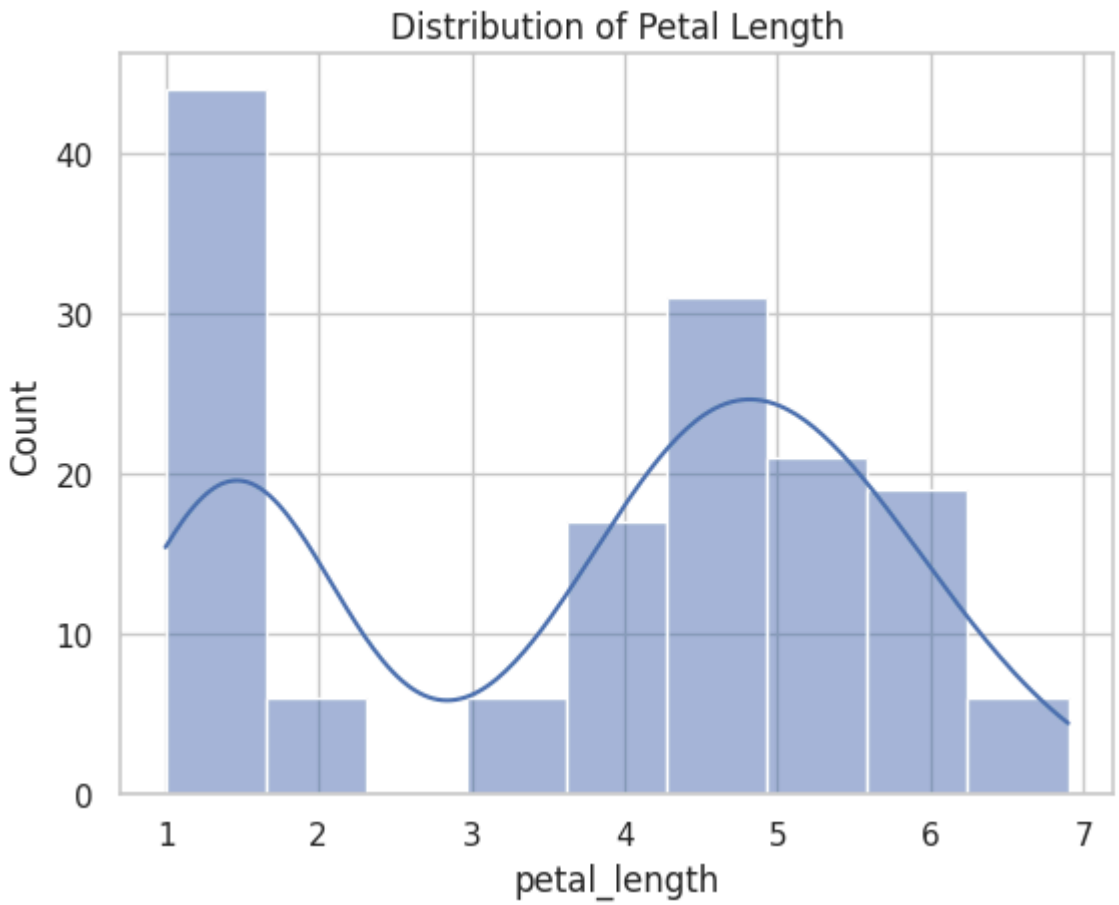
```
df_iris.describe()
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

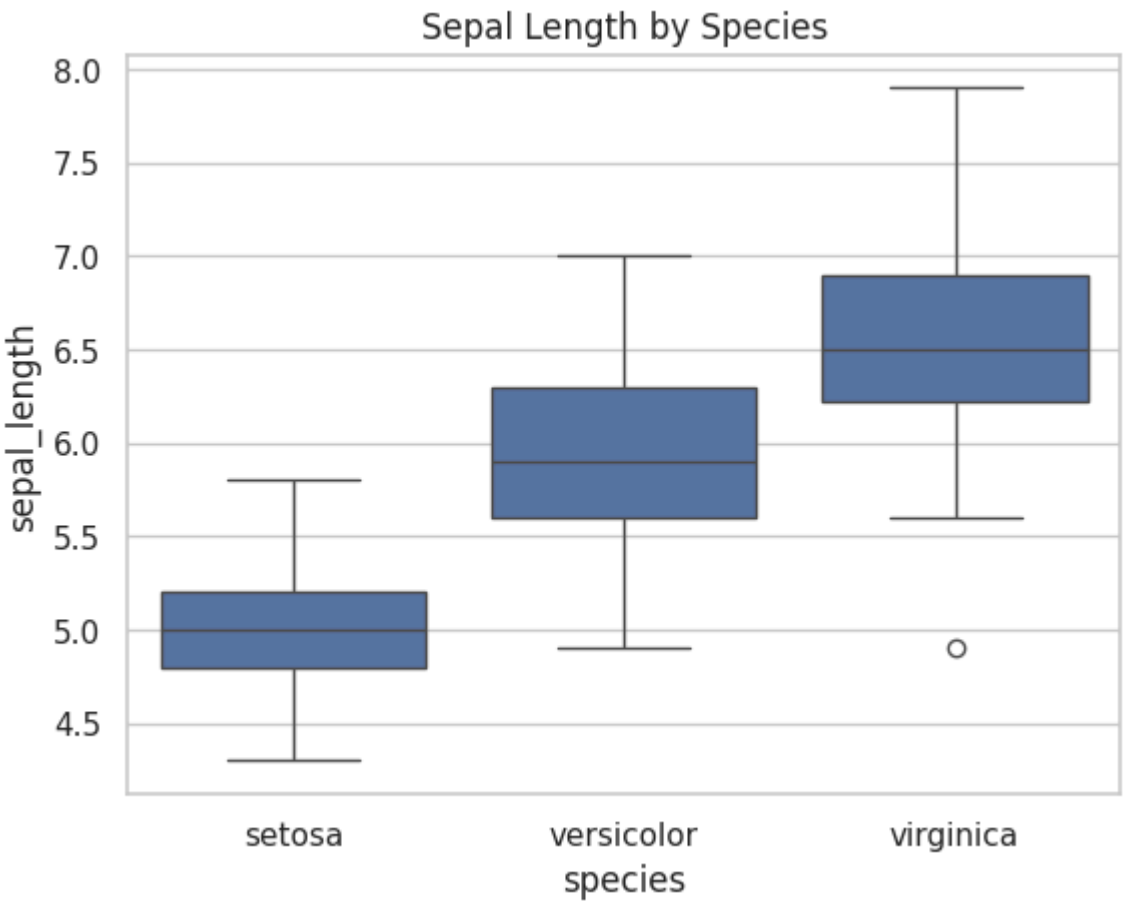
```
sns.histplot(df_iris["sepal_length"], kde=True)
plt.title("Distribution of Sepal Length")
plt.show()
```



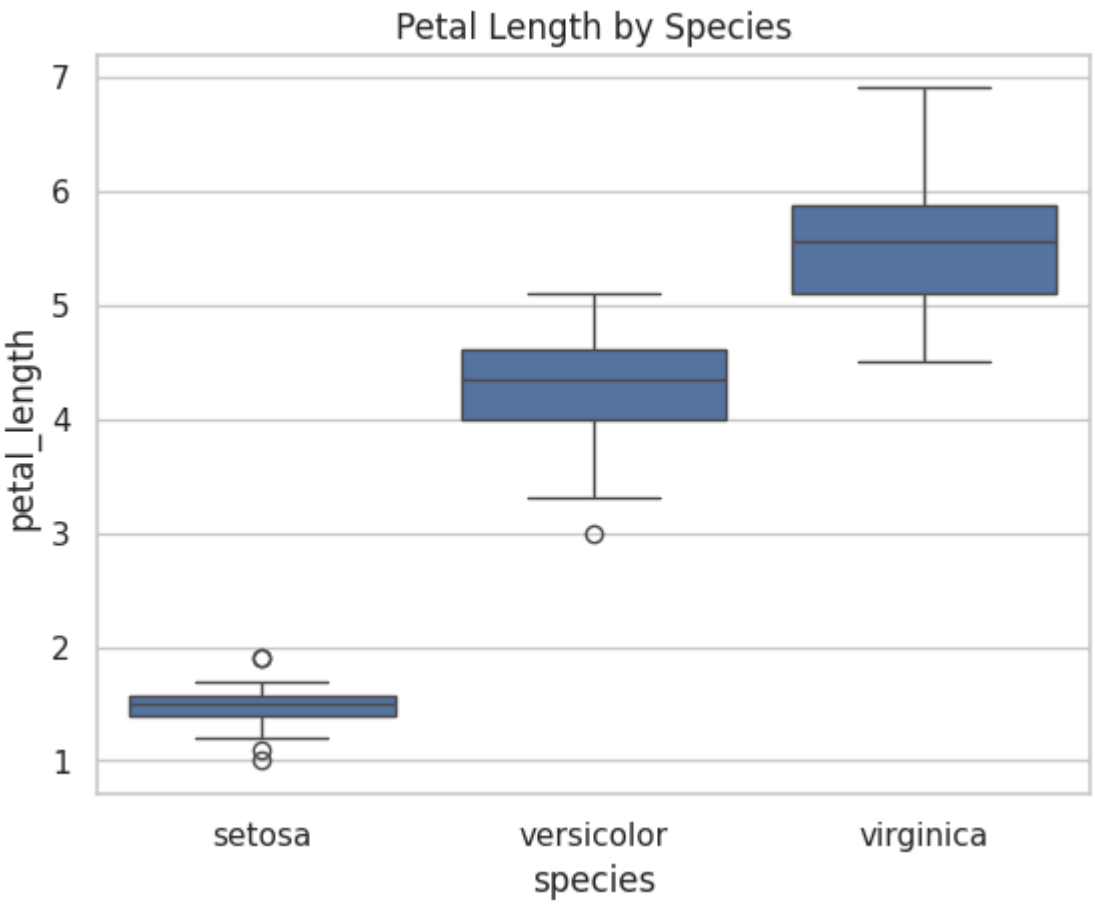

```
sns.histplot(df_iris["petal_length"], kde=True)
plt.title("Distribution of Petal Length")
plt.show()
```




```
sns.boxplot(x="species", y="sepal_length", data=df_iris)
plt.title("Sepal Length by Species")
plt.show()
```

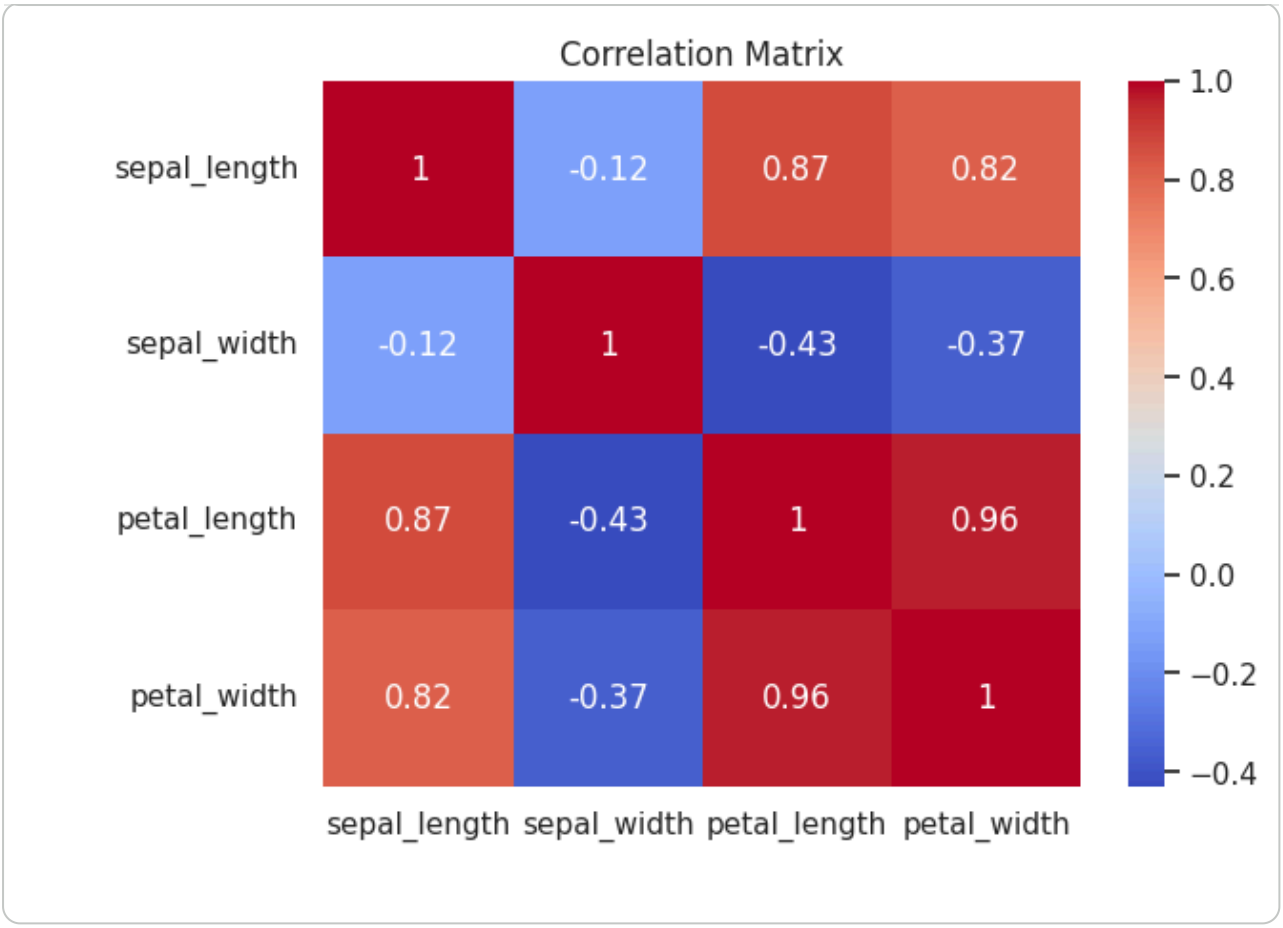



```
sns.boxplot(x="species", y="petal_length", data=df_iris)
plt.title("Petal Length by Species")
plt.show()
```

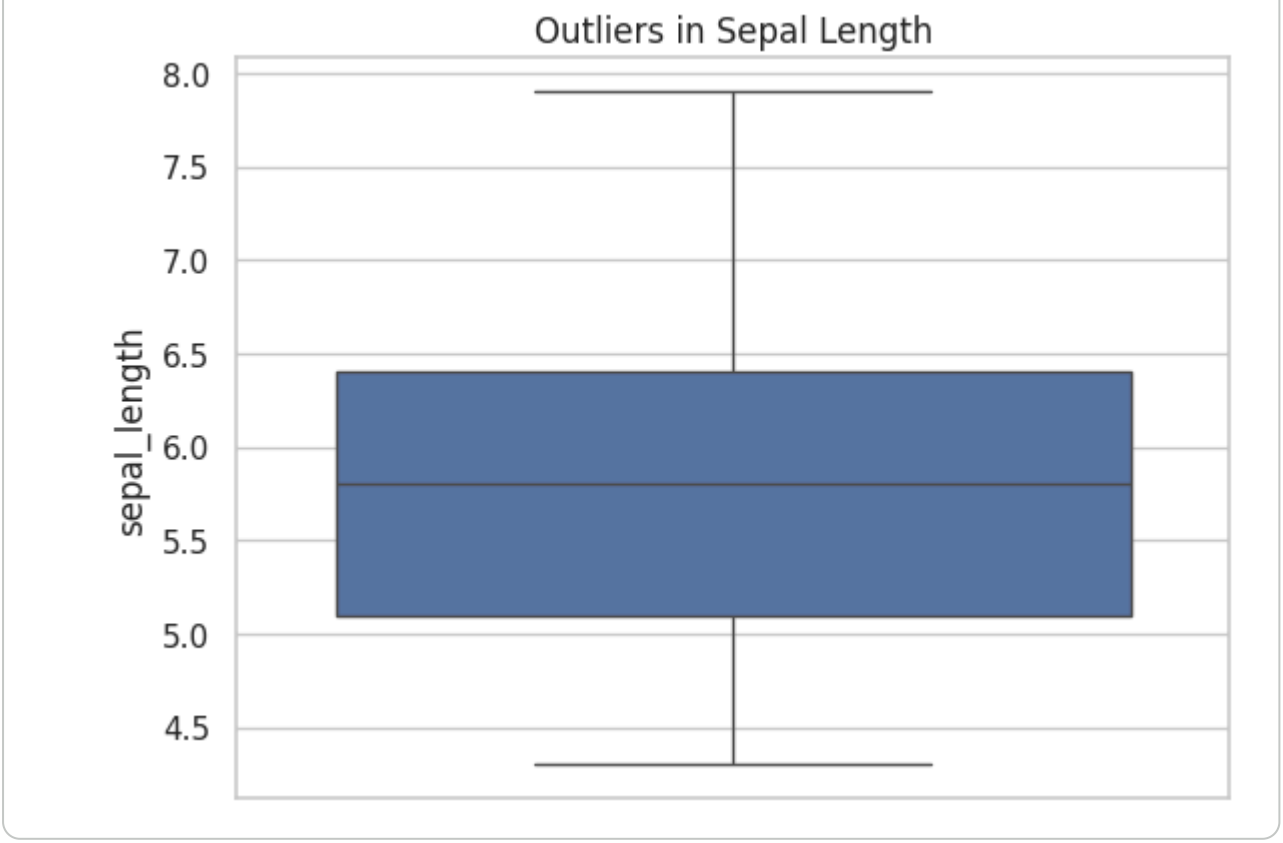



```
numeric_cols = ["sepal_length", "sepal_width", "petal_length", "petal_width"
corr = df_iris[numeric_cols].corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```
sns.boxplot(y=df_iris["sepal_length"])
plt.title("Outliers in Sepal Length")
plt.show()
```



The Iris dataset shows clear differences between species in both sepal and petal measurements. Petal length and petal width have a strong positive correlation. Sepal measurements vary less across species. The dataset is clean, with no missing values, and suitable for classification tasks

Dataset 4: Tips – EDA

```
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.cs
df_tips = pd.read_csv(url)
df_tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with df_tips | New interactive sheet

```
df_tips.shape
```

```
(244, 7)
```

```
df_tips.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    object
 3   smoker      244 non-null    object
 4   day         244 non-null    object
 5   time        244 non-null    object
 6   size        244 non-null    int64
dtypes: float64(2), int64(1), object(4)
memory usage: 13.5+ KB
```

```
df_tips.isna().sum()
```

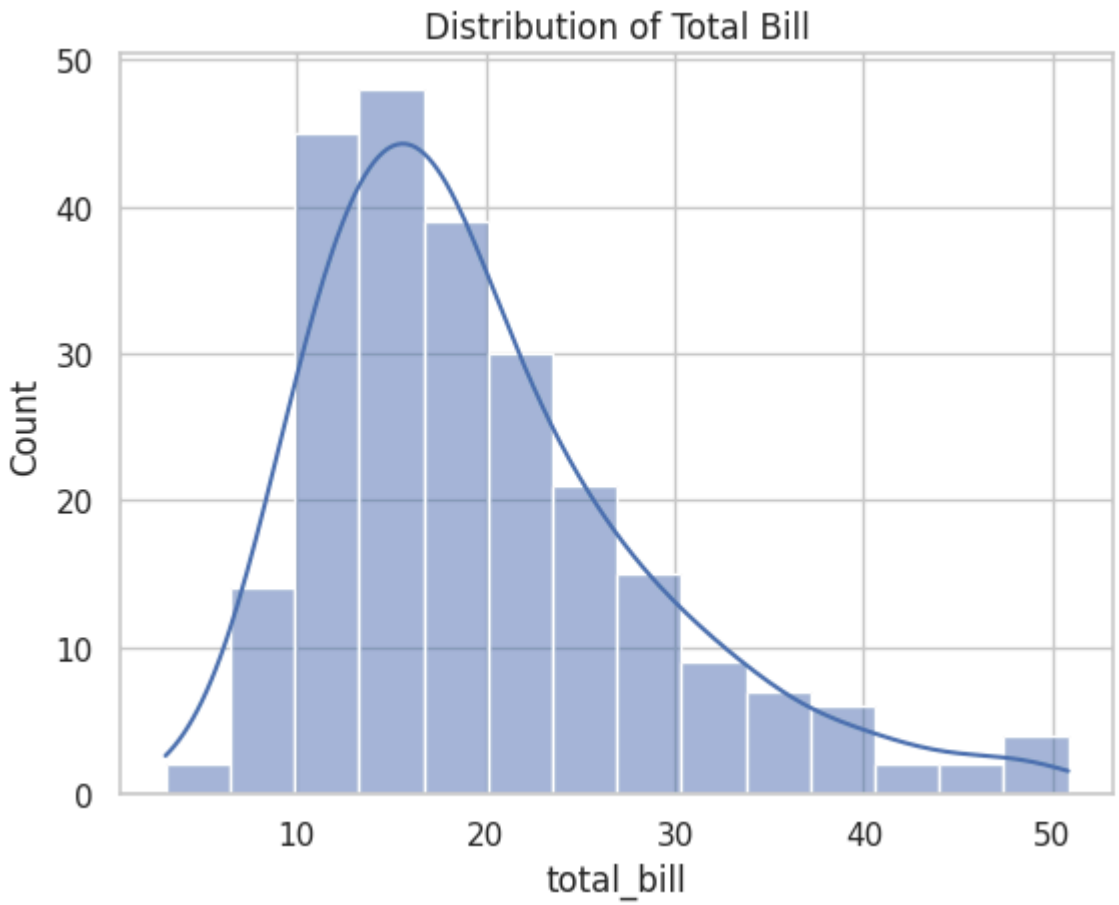
	0
total_bill	0
tip	0
sex	0
smoker	0
day	0
time	0
size	0

dtype: int64

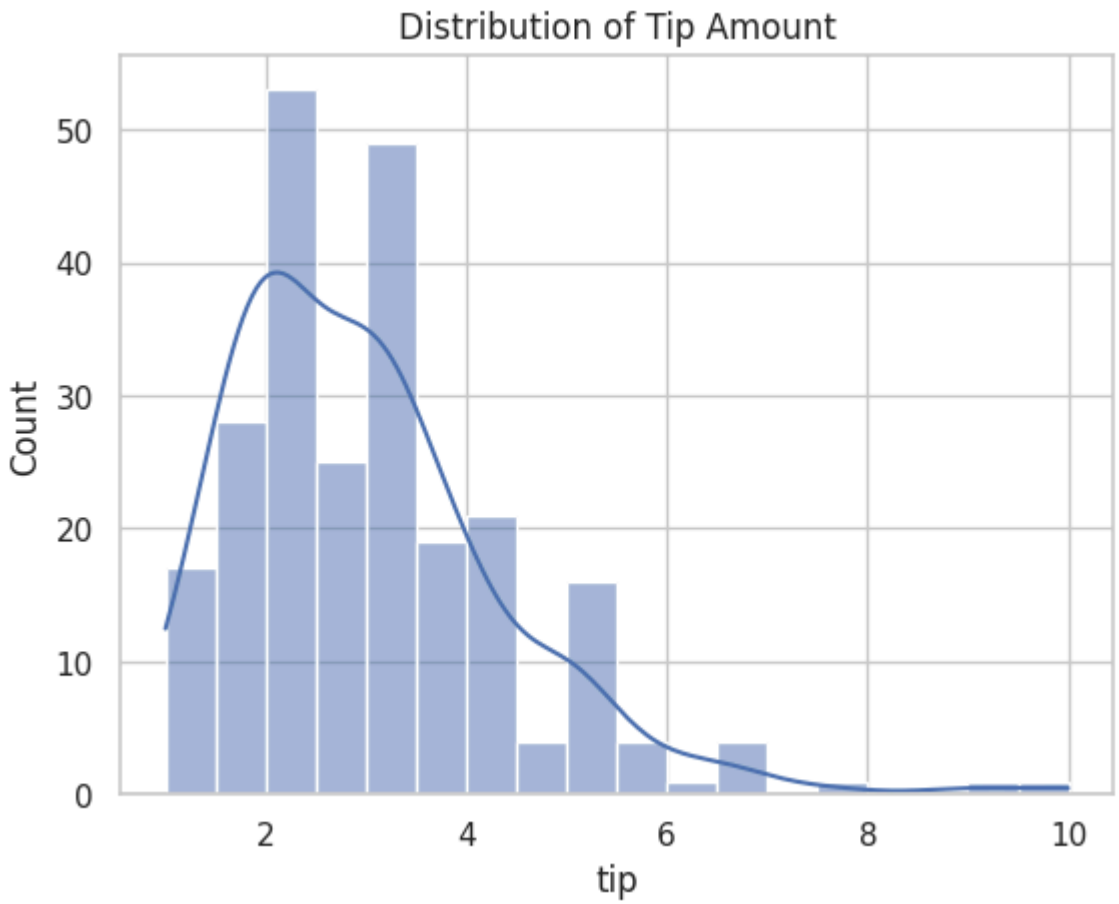
```
df_tips.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

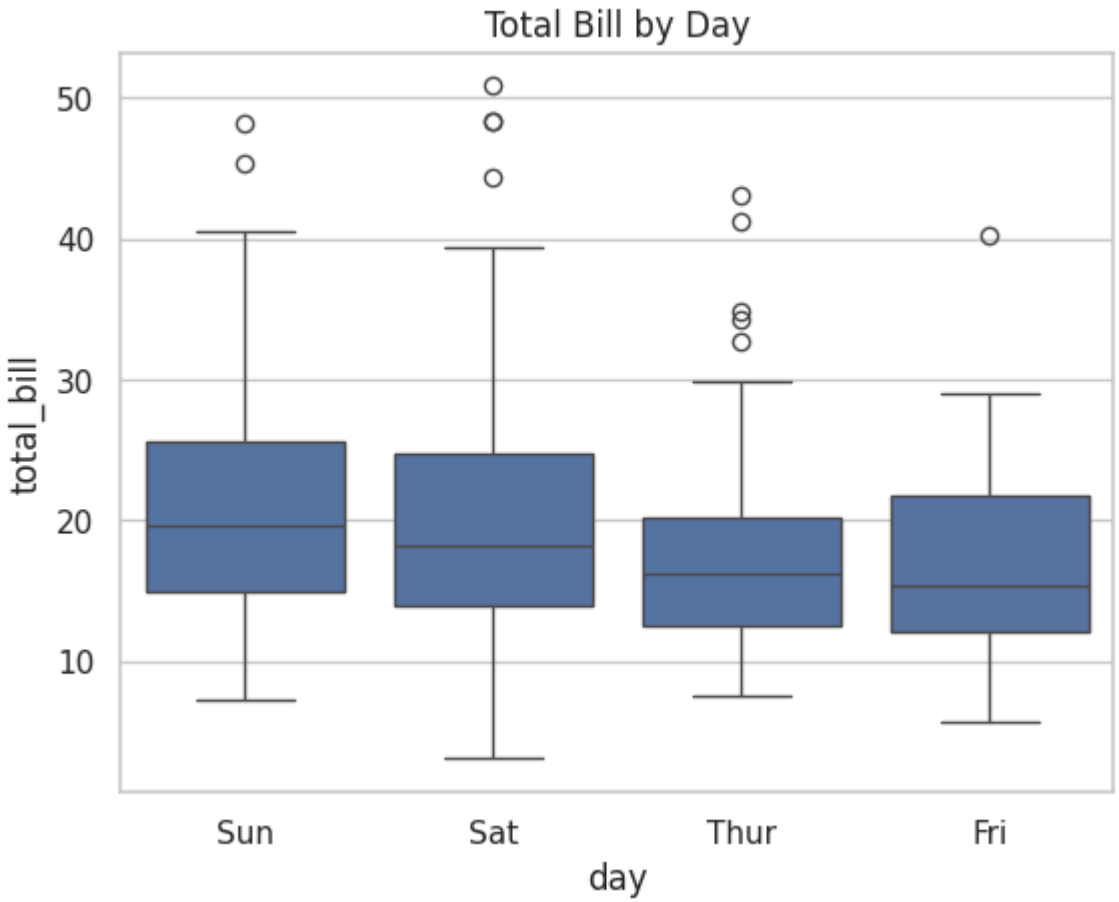
```
sns.histplot(df_tips["total_bill"], kde=True)
plt.title("Distribution of Total Bill")
plt.show()
```



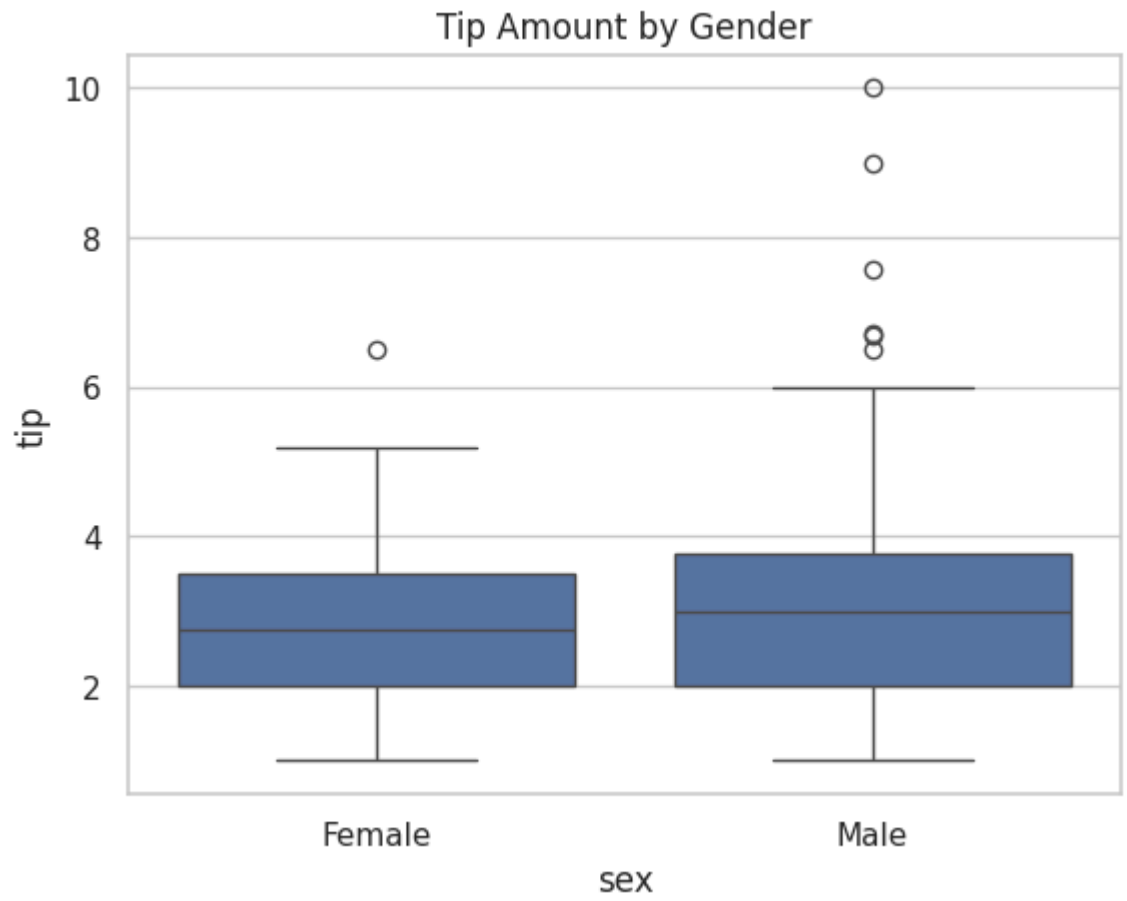

```
sns.histplot(df_tips["tip"], kde=True)
plt.title("Distribution of Tip Amount")
plt.show()
```




```
sns.boxplot(x="day", y="total_bill", data=df_tips)
plt.title("Total Bill by Day")
plt.show()
```



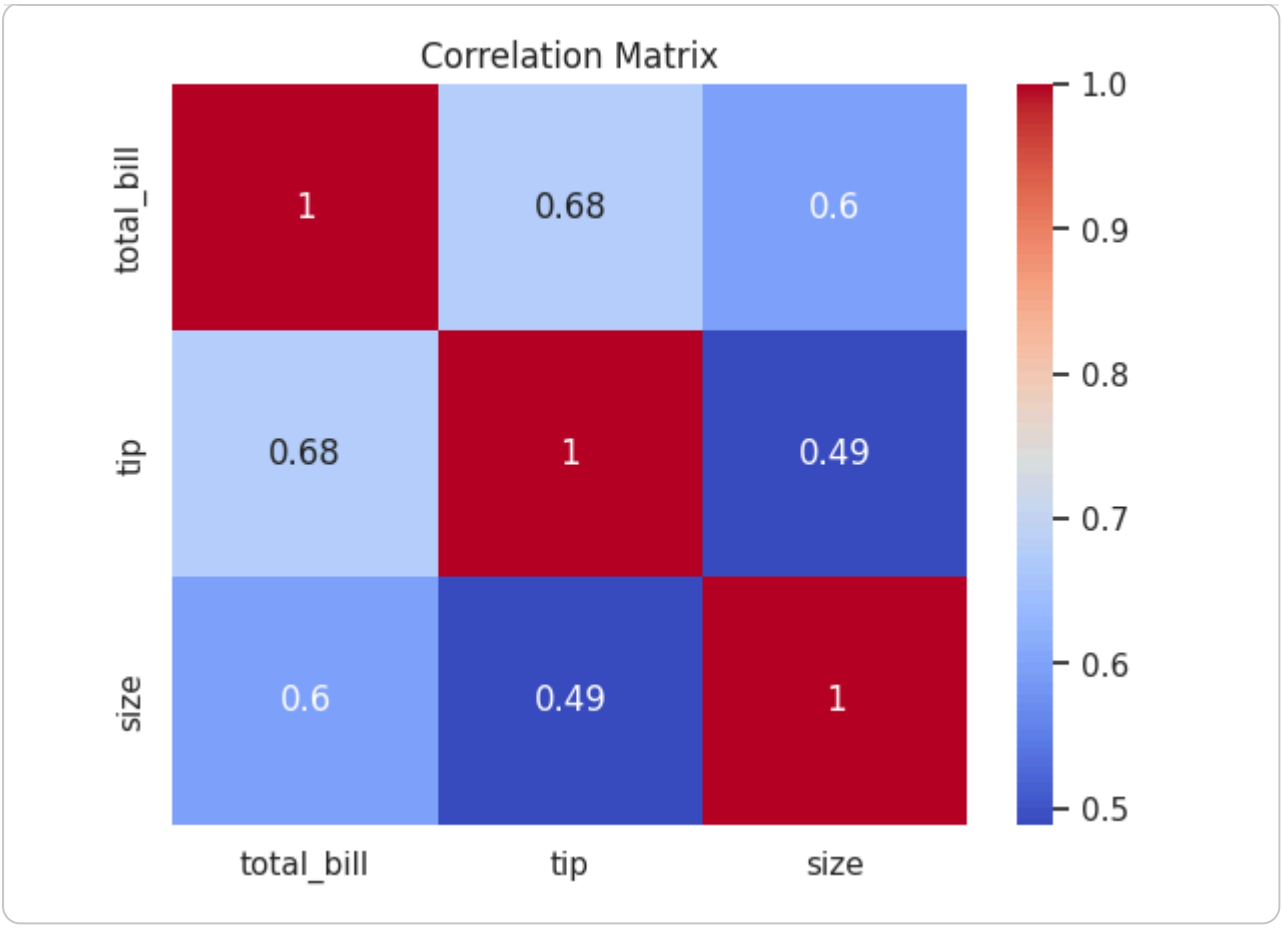

```
sns.boxplot(x="sex", y="tip", data=df_tips)
plt.title("Tip Amount by Gender")
plt.show()
```



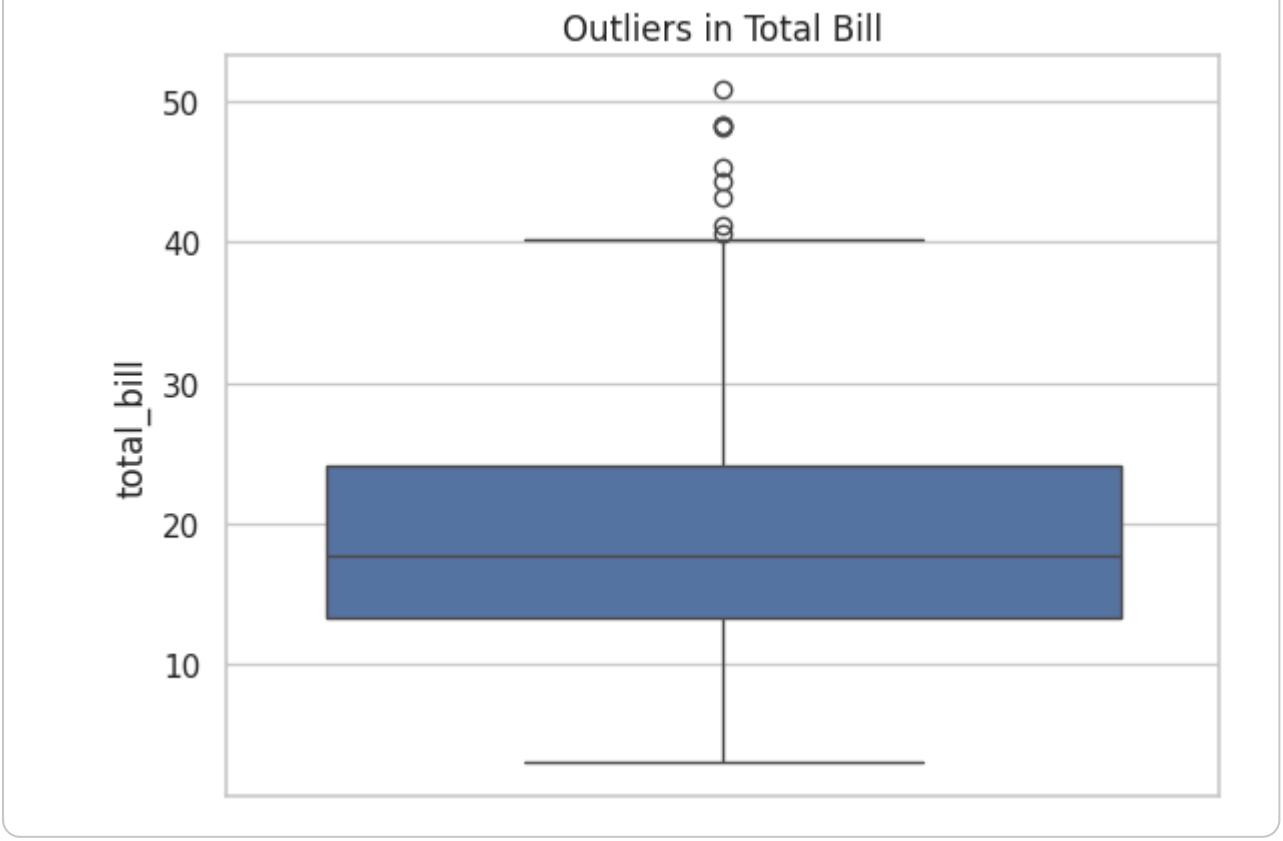
Start coding or generate with AI.

```
numeric_cols = ["total_bill", "tip", "size"]
corr = df_tips[numeric_cols].corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```
sns.boxplot(y=df_tips["total_bill"])
plt.title("Outliers in Total Bill")
plt.show()
```



The Tips dataset shows a positive correlation between total bill and tip amount. Larger groups tend to have higher bills. Tip amounts vary across days and between genders. The dataset is clean and suitable for simple regression or classification tasks

Close

```
Dataset 5: Wine Quality – EDA
```

Dataset 5: Wine Quality – EDA

```
url = "https://raw.githubusercontent.com/plotly/datasets/master/winequality-re
df_wine = pd.read_csv(url)
df_wine.head()
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978

Next steps: Generate code with df_wine | New interactive sheet

```
df_wine.shape
```

```
(1599, 12)
```

```
df_wine.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   fixed acidity         1599 non-null   float64
 1   volatile acidity      1599 non-null   float64
 2   citric acid           1599 non-null   float64
 3   residual sugar        1599 non-null   float64
 4   chlorides             1599 non-null   float64
 5   free sulfur dioxide   1599 non-null   float64
 6   total sulfur dioxide  1599 non-null   float64
 7   density               1599 non-null   float64
 8   pH                    1599 non-null   float64
```

```
 9   sulphates             1599 non-null   float64
 10  alcohol               1599 non-null   float64
 11  quality               1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
```